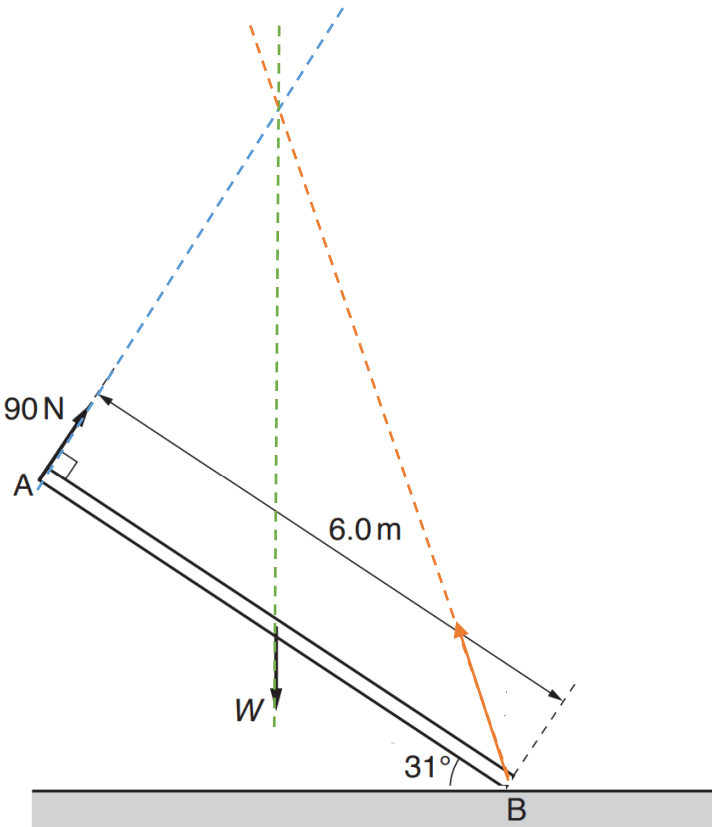


1	(a)(i)	Using $v = u + at$ for vertical motion, $5.5\sin(33^\circ) = -5.5\sin(33^\circ) + (9.81)t$ $t = 0.61 \text{ s}$	C1 A1
	(a)(ii)	Horizontal distance $= 5.5\cos(33^\circ) \times 0.61$ $= 2.8 \text{ m}$	C1 A1
	(a)(iii)	Ratio $= [5.5\cos(33^\circ)]^2/[5.5]^2$ $= 0.70$	C1 A1
		Marker's comment:	
	(b) (i)	 Students must show that the lines of action of all 3 lines intersect at a common point.	B1
	(b)(ii)	Taking moment about B, $90 \times 6.0 = W \times 3.0 \cos(31^\circ)$ $W = 210 \text{ N}$	C1 A1

